Chapter 11. Cluster Analysis

K-means, K-medoids, hierarchical clustering, Gaussian mixtures and the EM algorithm, choosing K

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Unlike classification (Ch. 10), clustering groups similar observations without labels (unsupervised learning). A clustering is an encoder $C : \{1, \dots, n\} \to \{1, \dots, K\}$.

1 Dissimilarity and Within-Cluster Scatter

Definition 11.1: within-cluster scatter

For a dissimilarity $d(i, j) \geq 0$,

$$W(C) = \frac{1}{2} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i:C(i)=k} \sum_{j:C(j)=k} d(i, j), \qquad n_k = \#\{i : C(i) = k\}.$$

Smaller W is better.

Theorem 11.2: with squared Euclidean distance $W(C)$ is the within-cluster sum of squares

If $d(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, then

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2, \qquad \bar{\mathbf{x}}_{(k)} = \frac{1}{n_k} \sum_{C(i)=k} \mathbf{x}_i,$$

and $T = \sum_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 = W(C) + B(C)$ with $B(C) = \sum_k n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$.

Proof. Within cluster k, $\mathbf{x}_i - \mathbf{x}_j = (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) - (\mathbf{x}_j - \bar{\mathbf{x}}_{(k)})$ and $\sum_{C(i)=k} (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) = \mathbf{0}$, so

$$\sum_{C(i)=k} \sum_{C(j)=k} \|\mathbf{x}_i - \mathbf{x}_j\|^2 = 2n_k \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2;$$

multiply by $\frac{1}{2} \cdot \frac{1}{n_k}$. $T = W + B$ is the trace of the MANOVA SSP decomposition (Ch. 6). ∎

Since T is fixed, minimizing W is maximizing B.

1.1 Limits of combinatorial algorithms

The number of ways to partition n items into K non-empty clusters (Stirling number of the second kind) is

$$A(n, K) = \frac{1}{K!} \sum_{k=1}^{K} (-1)^{K-k} \binom{K}{k} k^n \sim \frac{K^n}{K!}.$$

E.g. $A(19, 4) \approx 10^{10}$: exhaustive search is impossible, so greedy iterative algorithms are used.

2 K-means and K-medoids

2.1 K-means

With $d(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, any point $\boldsymbol{m}$ satisfies $\sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}\|^2 \geq \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, so

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2 = \min_{\boldsymbol{m}_1, \dots, \boldsymbol{m}_K} L(C \mid \boldsymbol{m}), \qquad L(C \mid \boldsymbol{m}) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}_k\|_2^2.$$

Definition 11.3: K-means algorithm

(i) Minimize over C with $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ fixed: update $C(i) = k$ for the closest center.

(ii) Minimize over $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ with C fixed: update $\boldsymbol{m}_k$ by the new average of cluster k.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Theorem 11.4: convergence

K-means stops after finitely many iterations at a local minimum.

Proof. Each step minimizes L in one block of variables with the other fixed, so L is non-increasing. There are finitely many encoders C and step (ii) is unique for a given C, so while L strictly decreases no C can recur; the algorithm stops in finitely many steps. ■

The result depends on the starting values, so run several starts and keep the smallest W. K-means is scale-dependent; consider standardizing.

2.2 K-medoids

Use an arbitrary dissimilarity and restrict centers to observations.

(i) Minimize over C: update $C(i) = k$ for the closest medoid $\boldsymbol{m}_k$.

(ii) Minimize over $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$: update $\boldsymbol{m}_k$ by the item in the kth cluster that minimizes the total dissimilarity to the cluster.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Using actual observations as representatives makes it robust to outliers and usable for categorical data.

3 Hierarchical Clustering

3.1 Agglomerative

Start with singletons and repeatedly merge the two closest clusters. Between-cluster distances (linkage):

Single linkage	$d(G_1, G_2) = \min_{i \in G_1, j \in G_2} d_{ij}$
Complete linkage	$d(G_1, G_2) = \max_{i \in G_1, j \in G_2} d_{ij}$
Average linkage	$d(G_1, G_2) = \text{average}_{i \in G_1, j \in G_2} d_{ij}$

Single linkage tends to produce long chained clusters, complete linkage compact round ones. The result is drawn as a dendrogram and cut at a chosen height.

Distances between observations:

$$p\text{-norm: } d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p = \Big(\sum_{l=1}^{d} |x_l - y_l|^p \Big)^{1/p},$$

$$\text{standardized distance: } d^2(\mathbf{x}, \mathbf{y}) = \sum_{l=1}^{d} \frac{(x_l - y_l)^2}{s_l^2} = (\mathbf{x} - \mathbf{y})^\top \mathbf{D}^{-1} (\mathbf{x} - \mathbf{y}).$$

3.2 Divisive

Start from a single cluster and split repeatedly (e.g. $K = 2$ means at each step).

4 Gaussian Mixtures and the EM Algorithm

4.1 Model

$\mathbf{X}_1, \dots, \mathbf{X}_n \overset{\text{iid}}{\sim} g(\mathbf{x})$ with latent labels $\delta_1, \dots, \delta_n \overset{\text{iid}}{\sim}$, $P(\delta_i = k) = \pi_k$, and $\mathbf{X} \mid \delta = k \sim N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$:

$$\mathbf{X} \sim g(\mathbf{x}) = \sum_{k=1}^{K} \operatorname{pdf}(\mathbf{x} \mid \delta = k) P(\delta = k) = \sum_{k=1}^{K} \pi_k g_k(\mathbf{x}),$$

$$L(\boldsymbol{\theta} \mid \mathbf{X}) = \prod_{i=1}^{n} g(\mathbf{x}_i) = \prod_{i=1}^{n} \Big\{ \sum_{k=1}^{K} g_k(\mathbf{x}_i) \pi_k \Big\},$$

giving $\hat{\pi}_k, \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k$. Then $\hat{P}(\delta = k \mid \mathbf{x}) = \hat{g}_k(\mathbf{x}) \hat{P}(\delta = k) / \hat{g}(\mathbf{x})$ and $C : \mathcal{X} \to \{1, \dots, K\}$, $C(\mathbf{x}) = k$ iff $\hat{P}(\delta = k \mid \mathbf{x}) \geq \hat{P}(\delta = j \mid \mathbf{x})$ for all j (soft clustering). With all $\boldsymbol{\Sigma}_k = \sigma^2 \mathbf{I}$ and $\sigma \to 0$ this becomes K-means.

4.2 The EM algorithm ($K = 2$)

$\mathbf{X}_1, \dots, \mathbf{X}_n \sim \pi_1 \phi(\mathbf{x}; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \pi_0 \phi(\mathbf{x}; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, $\pi_0 = 1 - \pi_1$. The observed-data log-likelihood

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log \{ \pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) + \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) \}$$

has no closed-form maximizer (log of a sum). If $\delta_1, \dots, \delta_n \in \{0, 1\}$ were observed, $L(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \prod_i \operatorname{pdf}(\mathbf{x}_i, \delta_i \mid \boldsymbol{\theta}) = \prod_i \operatorname{pdf}(\mathbf{x}_i \mid \delta_i, \boldsymbol{\theta}) \operatorname{pdf}(\delta_i)$ and

$$\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \sum_{i:\delta_i = 1} \log \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \sum_{i:\delta_i = 0} \log \pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$$

$$= \sum_{i=1}^{n} \left[\delta_i \log \pi_1 \phi_1(\mathbf{x}_i) + (1 - \delta_i) \log \pi_0 \phi_0(\mathbf{x}_i) \right],$$

which separates and gives closed forms such as $\hat{\pi}_1^{\text{MLE}} = n_1 / n$.

Definition 11.5: EM algorithm

(i) $\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n)$ has an easy MLE, but $\delta_1, \dots, \delta_n$ are unobserved; (ii) iteratively approximate the maximum. At $\boldsymbol{\theta}^{(m)}$:

- **E-step**: predict δ_i by $\hat{\delta}_i = \mathrm{E}(\delta_i \mid \mathbf{x}_i, \boldsymbol{\theta}^{(m)})$, i.e. compute $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) := \mathrm{E}[\ell(\boldsymbol{\theta} \mid D) \mid D_{\text{obs}}, \boldsymbol{\theta}^{(m)}]$. Since the complete-data log-likelihood is linear in δ_i,

$$\hat{\delta}_i = \frac{\pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})}{\pi_0^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_0^{(m)}, \boldsymbol{\Sigma}_0^{(m)}) + \pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})} \quad \text{(responsibility)}.$$

- **M-step**: $\boldsymbol{\theta}^{(m+1)} = \arg\max_{\boldsymbol{\theta}} L(\boldsymbol{\theta} \mid \mathbf{X}^n, \hat{\boldsymbol{\delta}}^n)$, i.e. $\arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$:

$$\pi_1 = \frac{\sum \hat{\delta}_i}{n}, \quad \boldsymbol{\mu}_1 = \frac{\sum \hat{\delta}_i \mathbf{x}_i}{\sum \hat{\delta}_i}, \quad \boldsymbol{\Sigma}_1 = \frac{\sum \hat{\delta}_i (\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^\top}{\sum \hat{\delta}_i},$$

and similarly for component 0 with weights $1 - \hat{\delta}_i$.

Repeat until $d(\boldsymbol{\theta}^{(m)}, \boldsymbol{\theta}^{(m+1)})$ is small.

Theorem 11.6: monotonicity of EM

$\ell(\boldsymbol{\theta}^{(m+1)}) \geq \ell(\boldsymbol{\theta}^{(m)})$.

Proof. $\ell(\boldsymbol{\theta}) = Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$ with $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \mathrm{E}[\log p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta}) \mid \mathbf{X}, \boldsymbol{\theta}^{(m)}]$. By Gibbs' (Jensen's) inequality $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) \leq H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})$ (KL divergence ≥ 0). The M-step does not decrease Q, so $\ell(\boldsymbol{\theta}^{(m+1)}) - \ell(\boldsymbol{\theta}^{(m)}) = \{Q(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - Q(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} - \{H(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} \geq 0$. ■

EM may converge to a local maximum (or saddle point), and the likelihood is unbounded as some $\boldsymbol{\Sigma}_k$ becomes singular, so use several starts and regularization.

5 Choosing K and Other Topics

- **Scatter decomposition**: $W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, $B(C) = \sum_{k=1}^{K} n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$. W always decreases as K grows, so it cannot choose K by itself (elbow method: where the decrease flattens).
- Indices: $\mathrm{CI}(K, c) = W(K, c)/T$ (smaller is better) and Calinski–Harabasz $\mathrm{CH}(K, c) = \dfrac{B(K, c)/(K-1)}{W(K, c)/(n-K)}$ (larger is better; the analogue of the one-way MANOVA F).
- **Gap statistic**: compare with the expected $\log W_K^*$ under a reference distribution (uniform over the data range),

$$\mathrm{Gap}(K) = \mathrm{E}^*[\log W_K^*] - \log W_K, \qquad s(K) = \text{(adjusted) s.d. of } \log W_K^*,$$

  and choose the smallest K with $\mathrm{Gap}(K) \geq \mathrm{Gap}(K+1) - s(K+1)$.
- **Clustering variables**: define $d(v_i, v_j) = 1 - r_{ij}$ or $1 - r_{ij}^2$ and apply hierarchical clustering.